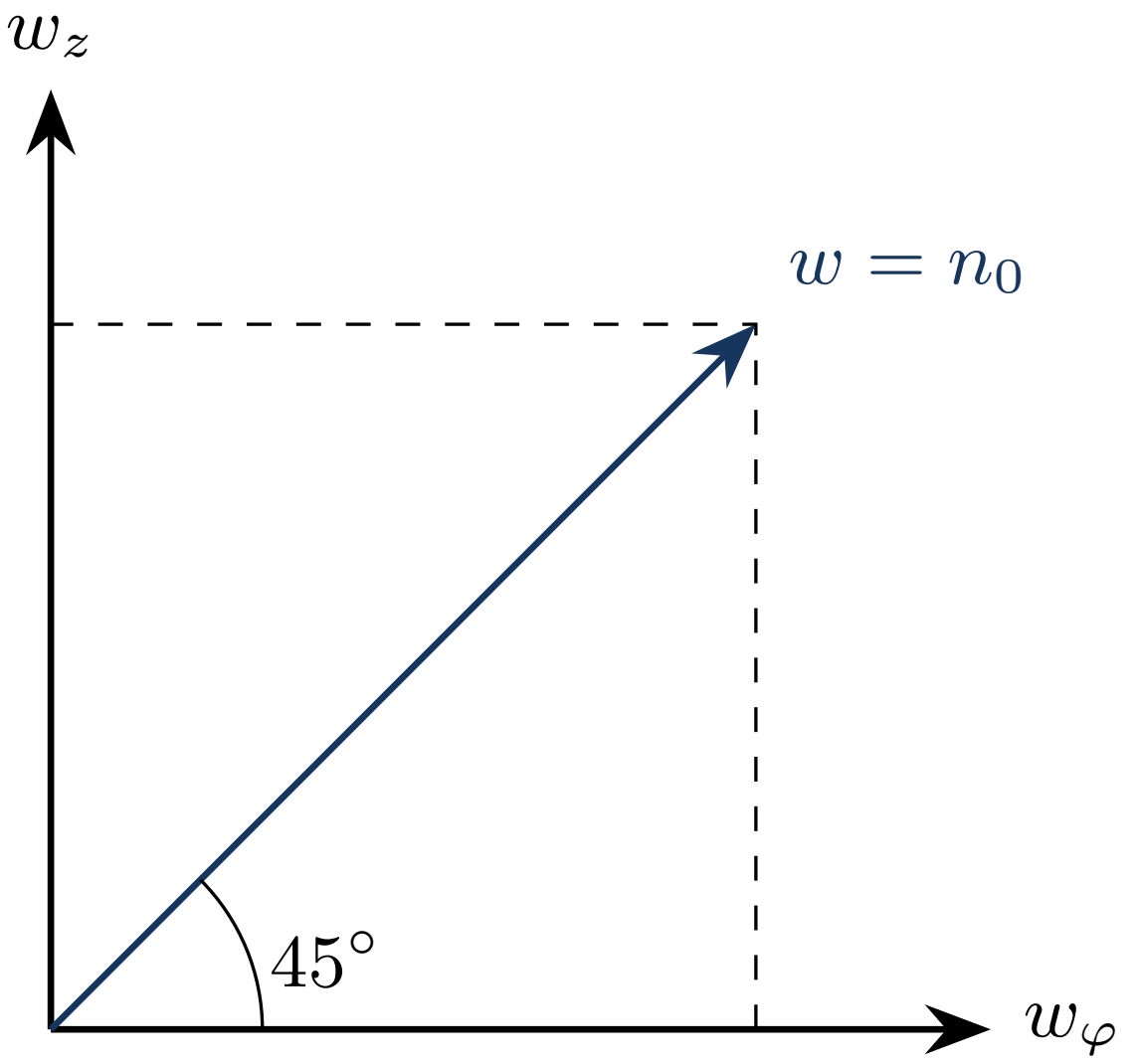

$$w_\varphi^2/r_0 = a/(2r_0^2)$$

$$w_z = w_\varphi$$